QUIZ 1 -MTH301-page5-8

Graded

Student

YATIN PREETAM BHOJWANI

Total Points

4 / 10 pts

Question 1

Q3

0 / 5 pts

✔ **+ 0 pts** Completely wrong/not attempted

Question 2

Q4

4 / 5 pts

✔ **+ 4 pts** partial correct

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3. Let (M, d) be a metric space. A point $x \in M$ is said to be a **Boundary Point** of a set A iff $B_\epsilon(x) \cap A \neq \emptyset$ and $B_\epsilon(x) \cap A^C \neq \emptyset$ for every $\epsilon > 0$.

If F is a nonempty bounded subset of $\mathbb{R}$ with usual metric, show that $\sup F$ is a boundary point of F.

[5]

4. Consider $M = [0, 2)$ and define $d_1(x, y) = \min\{|x - y|, 0.1\}$.

- Determine all open balls in (M, d_1).
- Are $\{0\}$ and $\{1\}$ an open set or a closed set in (M, d_1)? Justify.
- Is $(0, 0.1]$ an open set in (M, d_1)? Justify.

[2+2+1]

• $B_r(x) = \{y : d(y,x) < r\}$

for $r > 0.1$.

$d(y,x) = \min\{|y-x|, 0.1\}$
$\leq 0.1 < r$.

$\Rightarrow$ for $r > 0.1$ and $x \in M$.

$B_r(x)$ is the set M.

2

for $r \leq 0.1$

$d(y,x) < r \leq 0.1$
$\Rightarrow d(y,x) < 0.1$
$\Rightarrow d(y,x) = |y-x|$

the standard metric on $\mathbb{R} \setminus \{0\}$

consider any $x \in M$ and $r \leq 0.1$

$B_r(x)$ is the set of pts in $(x-r, x+r)$ if $x-r > 0$ and $x+r < 2$ $\cap M$

it is $[0, x+r)$ if $x-r < 0$ and $x+r < 2$

it is $(x-r, 2)$ if $x-r > 0$ and $x+r > 2$

it is $[0,2)$ if $x-r < 0$ and $x+r > 2$.

second part.

~~0 is closed.~~

Claim: $\{0\}$ is closed.

i will show this by showing $(0,2)$ is open

consider $x \in (0,2)$

i can find $c, K \in \mathbb{R}$

s.t $0 < c < x < K < 2$ and $K-x < 0.1$

and $x - c \leq 0.1$

$\epsilon := \min\{x-c, K-x\} < 0.1$

$\Rightarrow$ ~~$< |y - x|$~~ take set

$= [\min\{x - r, 0$

take $\varepsilon :=$

~~for $\forall$ y~~ $\in$ ~~$(x-\varepsilon, x-\varepsilon, x+\varepsilon\}$~~

$d(y, x)$ ~~$= |y-x|$~~ ~~$< \varepsilon$.~~ (as $\varepsilon < 0.1\}$

$\Rightarrow$ $|y$

as ~~$\varepsilon < 0.1$~~

~~$|y-x| < 0.1$~~

~~$\Rightarrow d(y,x) = |y-x|$~~

consider the ball

$B_{\varepsilon}(x) = \{ y : d(y,x) < \varepsilon \}$

~~then as $\varepsilon < 0.1$ the metric~~

we know $|y-x| < \varepsilon < 0.1$

$\exists$ ~~$d(y,x)$~~

Part 3:

$(0, 0.1]$ is not open ball.

I will show that $\exists$ no ε

such that $B_{\varepsilon}(0.1) \subseteq (0, 0.1)$

for $\varepsilon > 0.1$ we know

$B_{\varepsilon} = M$ which

~~does not belong to $(0, 0.1]$~~

and $M \notin (0, 0.1]$